

MATHEMATICAL ANALYSIS 1
HOMEWORK 3

- (1) (a) Let $X = \{1, 2, 3\}$ and $Y = \{a, b\}$.
- (i) Give an example of a function $f : X \rightarrow Y$ with $\text{dom}(f) = X$ that is surjective.
 - (ii) Explain why no function $f : X \rightarrow Y$ with $\text{dom}(f) = X$ can be injective.
 - (iii) Give an example of a function $f : X \rightarrow Y$ that is injective. What is the largest possible number of elements of $\text{dom}(f)$ for such an f ? Can such an f also be surjective?
- (b) Determine whether each of the following functions is injective, surjective, or bijective (explain your answers, either using the definitions or by sketching the graph and considering horizontal lines):
- (i) $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$
 - (ii) $f : [0, +\infty) \rightarrow \mathbb{R}, f(x) = x^2$
 - (iii) $f : [0, +\infty) \rightarrow [0, +\infty), f(x) = x^2$
 - (iv) $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^3$
- (2) Let $f(x) = \frac{2x+1}{x-3}$, with $\text{dom}(f) = \mathbb{R} \setminus \{3\}$.
- (a) Show that f is injective.
 - (b) Determine $\text{im}(f)$.
 - (c) Find the inverse function f^{-1} , and state $\text{dom}(f^{-1})$ and $\text{im}(f^{-1})$.
 - (d) Verify that $f(f^{-1}(y)) = y$ for all $y \in \text{dom}(f^{-1})$, and that $f^{-1}(f(x)) = x$ for all $x \in \text{dom}(f)$.
- (3) Prove the following lemma:

Lemma. *Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be monotonically increasing on some $A \subseteq \mathbb{R}$. Then $f + g$ is also monotonically increasing on A . If either f or g are strictly increasing on A , then so is $f + g$. The same statements hold if we replace everywhere the word ‘increasing’ with the word ‘decreasing’.*

- (4) Show that if $f : A \subseteq \mathbb{R} \rightarrow \mathbb{R}$ is monotone increasing on A , then $-f$ is monotone decreasing on A .
- (5) For the following pairs of functions f and g , compute $f \circ g$ and $g \circ f$, and determine their domains:
- (a) $f(x) = x^2, g(x) = x + 1$
 - (b) $f(x) = \sqrt{x}, g(x) = x - 2$
 - (c) $f(x) = \frac{1}{x}, g(x) = x^2 + 1$
 - (d) $f(x) = \sin(x), g(x) = 2x$
- (6) Prove the following lemma:

Lemma. *Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$. Then:*

$$\begin{array}{lll} f, g \text{ are both monotone increasing} & \Rightarrow & g \circ f \text{ is monotone increasing.} \\ f, g \text{ are both monotone decreasing} & \Rightarrow & g \circ f \text{ is monotone increasing.} \\ f, g \text{ are monotone of different kinds} & \Rightarrow & g \circ f \text{ is monotone decreasing.} \end{array}$$

- (7) Let t_c and s_c be the translation and scaling functions, respectively. Let r be the reflection function. Consider the function $f(x) = x^3$ restricted to the interval $[-1, 1]$. Sketch the following:
- (a) $f \circ t_c$ for $c = -1, 0, 1$
 - (b) $t_c \circ f$ for $c = -1, 0, 1$
 - (c) $f \circ s_c$ for $c = \frac{1}{2}, 2$
 - (d) $s_c \circ f$ for $c = \frac{1}{2}, 2$
 - (e) The difference $f \circ r - r \circ f$.
- (8) Sketch the graphs of the following functions (choose a relevant part of the graph to sketch):
- (a) $\sin(x), \sin(2x), \sin(-x)$
 - (b) $\log_{\frac{1}{5}} x, \log_5 x, \log_5(x^2)$

- (c) $\cos(x), \cos(2x), \cos(x - \frac{\pi}{2})$
- (9) For the following functions f and subsets $A \subseteq \text{dom}(f)$, determine $\sup_A f$, $\inf_A f$, and whether the maximum and minimum are attained on A :
- (a) $f(x) = x^2, A = [0, 2]$
 - (b) $f(x) = x^2, A = (0, 2)$
 - (c) $f(x) = x^3 + x^2 + 2x + 1, A = [0, 2]$
 - (d) $f(x) = \frac{1}{x}, A = (0, 1]$
 - (e) $f(x) = \frac{1}{x}, A = [1, +\infty)$
 - (f) $f(x) = e^x, A = [-1, 1]$
 - (g) $f(x) = \frac{x}{1+x}, A = [0, +\infty)$
 - (h) $f(x) = \sin(x), A = (0, \pi)$
 - (i) $f(x) = \arctan(x), A = [0, +\infty)$
- For parts (g)–(i), explain your answers using the definitions of supremum and infimum.
- (10) Using trigonometric identities, simplify the following expressions:
- (a) $\frac{\sin(x)}{\cos(x)} + \frac{\cos(x)}{\sin(x)}$
 - (b) $(\sin(x) + \cos(x))^2 + (\sin(x) - \cos(x))^2$
 - (c) $\frac{1 - \cos(2x)}{\sin(2x)}$
- (11) Prove the following equalities:
- (a) $\tan(x) + \cot(x) = \frac{2}{\sin(2x)}$
 - (b) $\cos(3x) = 4\cos^3(x) - 3\cos(x)$

Additional Practice Problems

★★ These problems are for you to practice ★★

★★ They are not part of the homework assignment ★★

- (1) Sketch the graphs of the following functions (choose a relevant part of the graph to sketch):
 - (a) $\sin(x + \frac{\pi}{2})$, $\sin(x) + 1$
 - (b) $\log_{10}(\frac{2}{x})$, $\ln(-x)$, $\ln(x^{-1})$
 - (c) $-\cos(x)$, $\cos(-x)$
 - (d) $\tan(x)$, $\tan(2x)$, $\tan(x + \frac{\pi}{4})$, $-\tan(x)$
 - (e) $\arcsin(x)$, $\arcsin(2x)$, $\arcsin(-x)$, $\arcsin(x) + \frac{\pi}{2}$
 - (f) $\arctan(x)$, $\arctan(2x)$, $\arctan(-x)$, $-\arctan(x)$
- (2) Sketch the graphs of the following power functions:
 - (a) $x^{1/2}$, $x^{1/3}$, $x^{1/4}$
 - (b) x^{-1} , x^{-2} , x^{-3}
 - (c) $x^{2/3}$, $x^{3/2}$, $x^{-1/2}$
 - (d) $(x-1)^{1/2}$, $(x+2)^{-1}$, $2x^3$
- (3) For the following pairs of functions f and g , compute $f \circ g$ and $g \circ f$, and determine their domains:
 - (a) $f(x) = e^x$, $g(x) = \ln(x)$
 - (b) $f(x) = |x|$, $g(x) = x - 3$
 - (c) $f(x) = \sqrt{x}$, $g(x) = x^2$
 - (d) $f(x) = \frac{1}{x-1}$, $g(x) = \frac{1}{x}$
 - (e) $f(x) = x^{1/3}$, $g(x) = x^3$
 - (f) $f(x) = x^2 + 1$, $g(x) = \sqrt{x-1}$
- (4) For the following functions f and subsets $A \subseteq \text{dom}(f)$, determine $\sup_A f$, $\inf_A f$, and whether the maximum and minimum are attained on A :
 - (a) $f(x) = e^x$, $A = (-1, 1)$
 - (b) $f(x) = \sqrt{4-x^2}$, $A = (-2, 2)$
 - (c) $f(x) = \arctan(x)$, $A = \mathbb{R}$
 - (d) $f(x) = \frac{1}{1-x}$, $A = [0, 1)$
 - (e) $f(x) = x(2-x)$, $A = (0, 2)$
 - (f) $f(x) = x^{1/3}$, $A = [-8, 8]$
 - (g) $f(x) = x^{-2}$, $A = [1, +\infty)$
- (5) Determine whether the following functions are even, odd, or neither:
 - (a) $f(x) = x^4 + x^2$
 - (b) $f(x) = x^5 - x^3$
 - (c) $f(x) = x^3 + x^2$
 - (d) $f(x) = \frac{x}{x^2+1}$
 - (e) $f(x) = x^{-2}$
 - (f) $f(x) = |x| + x$
- (6) Using trigonometric identities, simplify the following expressions:
 - (a) $\sin(x+y)\sin(x-y)$
 - (b) $\cos^4(x) - \sin^4(x)$
 - (c) $\frac{1+\tan^2(x)}{1+\cot^2(x)}$
- (7) Prove the following equality:
 - (a) $\frac{\sin(x)+\sin(3x)}{\cos(x)+\cos(3x)} = \tan(2x)$
- (8) Simplify the following expressions involving powers:
 - (a) $(x^3)^{1/2} \cdot x^{-1/2}$
 - (b) $\frac{x^{2/3} \cdot x^{1/2}}{x^{-1/6}}$
 - (c) $(x^{-2}y^3)^{-1/2}$
 - (d) $\left(\frac{x^{1/3}}{x^{-2/3}}\right)^3$